



Development of 5-year risk prediction models for incident dementia and mortality in a community-dwelling older Japanese population: The Japan Prospective Studies Collaboration for Aging and Dementia (JPSC-AD)

Xiangyin Meng · Takanori Honda · Tomoyuki Ohara · Yoshihiko Furuta · Mao Shibata · Jun Hata · Yasuyuki Taki · Tatsuya Mikami · Tetsuya Maeda · Kenjiro Ono · Masaru Mimura · Ritsuko Hanajima · Jun-ichi Iga · Minoru Takebayashi · Toshiharu Ninomiya  · On the behalf of the JPSC-AD study group

Received: 29 January 2026 / Accepted: 4 May 2026
© The Author(s) 2026

Abstract Improving cognitively healthy survival is important for achieving healthy aging. Therefore, it would be valuable to estimate the future risk of either incident dementia or death in community-dwelling older adults. This study aimed to develop a set of risk prediction models for either incident dementia or death that can be applied according to data availability across diverse clinical settings, using longitudinal data from community-dwelling older Japanese adults. A total of 8,334 participants aged ≥ 65 years were prospectively followed up from 2016 to 2023.

Logistic regression was used to develop multivariable prediction models. The developed models were translated into simplified scoring systems based on the β coefficients. The discrimination abilities of the models were assessed by C-statistics, and the calibration was assessed by calibration plots. During the follow-up period, 1,151 participants developed either dementia or death. In the multivariable model using potential predictors commonly available in the primary care setting, age, male sex, formal education ≤ 9 years, diabetes mellitus, use of lipid-lowering agents, leanness, history of stroke, history of heart disease, history of respiratory disease, history of cancer, current smoking, no regular exercise habit, and

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s11357-026-02304-w>.

X. Meng · T. Ohara
Department of Neuropsychiatry, Graduate School of Medical Sciences, Kyushu University, Fukuoka, Japan

X. Meng · T. Honda · T. Ohara · Y. Furuta · M. Shibata · T. Ninomiya (✉)
Department of Epidemiology and Public Health, Graduate School of Medical Sciences, Kyushu University, Fukuoka, Japan
e-mail: ninomiya.toshiharu.734@m.kyushu-u.ac.jp

M. Shibata · J. Hata · T. Ninomiya
Department of Center for Cohort Studies, Graduate School of Medical Sciences, Kyushu University, Fukuoka, Japan

J. Hata
Department of Health Care Administration and Management, Graduate School of Medical Sciences, Kyushu University, Fukuoka, Japan

Y. Taki
Department of Nuclear Medicine and Radiology, Institute of Development, Aging and Cancer, Tohoku University, Sendai, Japan

T. Mikami
Department of Preemptive Medicine, Graduate School of Medicine, Hirosaki University, Hirosaki, Japan

T. Maeda
Division of Neurology and Gerontology, Department of Internal Medicine, School of Medicine, Iwate Medical University, Morioka, Iwate, Japan

K. Ono
Department of Neurology, Kanazawa University Graduate School of Medical Sciences, Kanazawa University, Kanazawa, Japan

low frequency of social interactions were selected as predictors. Adding cognitive status, depressive symptoms, apolipoprotein E-ε4 carrier status, and brain MRI markers to the model further enhanced its predictive performance. The developed models and simplified scores showed good discrimination and calibration. Risk stratification with our models may be useful for assessing the risks of incident dementia and death and, consequently, for prolonging cognitively healthy survival.

Keywords Risk prediction · Survival · Dementia · Public Health

Introduction

As life expectancy increases worldwide, population aging and the rising number of individuals living with dementia have become major public health concerns [1, 2]. Dementia is a leading cause of premature death and disability [3, 4], imposing enormous social and economic burdens [5]. Accordingly, public health and policy priorities have shifted from merely extending life expectancy to prolonging independent, cognitively healthy survival—that is, “healthy aging” [6, 7]. To achieve this goal, it is important to reduce not only mortality risk but also the risk of developing dementia. Notably, recent evidence indicates that cardiovascular risk-reduction and dementia-prevention strategies are often cost-effective [8, 9], further strengthening the case for timely intervention. Therefore, early identification of high-risk individuals

and management of risk factors are important public health challenges for prolonging cognitively healthy survival.

A prediction tool may be useful to stratify individuals according to their future risk of incident dementia and death, and to encourage practitioners to offer appropriate investigations and treatments. Numerous models have been developed to predict the risk of incident dementia or mortality separately [10, 11]. However, to our knowledge, few prediction models have been developed to estimate the risk of the composite endpoint of incident dementia or death. In addition, existing prediction models are not always feasible in routine practice [12], highlighting a tradeoff between feasibility and predictive performance. Prediction models based on easily accessible information in the primary care setting are highly practical, whereas including more resource-intensive variables, such as cognitive batteries or brain magnetic resonance imaging (MRI), may enhance predictive performance [13]. Therefore, it is important to develop a set of prediction models that can be tailored to various clinical settings, depending on the information available in each setting.

The Japan Prospective Studies Collaboration for Aging and Dementia (JPSC-AD) is a multisite, community-based prospective cohort study of dementia. The aims of the present study were to develop, using follow-up data from JPSC-AD, three risk prediction models for the composite endpoint of incident dementia or death for use in settings with different levels of available information, and to evaluate their performance in a cohort of community-dwelling older Japanese adults.

Methods

Study population

The JPSC-AD is a multisite, population-based prospective cohort study of dementia involving eight sites in Japan: Hirosaki City (Aomori Prefecture), Yahaba Town (Iwate Prefecture), Nakajima Town in Nanao City (Ishikawa Prefecture), Arakawa Ward (Tokyo), Ama Town (Shimane Prefecture), Nakayama Town in Iyo City (Ehime Prefecture), Hisayama Town (Fukuoka Prefecture), and Arao City (Kumamoto Prefecture). The study protocol and standardized

M. Mimura

Center for Preventive Medicine, Keio University School of Medicine, Tokyo, Japan

R. Hanajima

Division of Neurology, Department of Brain and Neurosciences, Faculty of Medicine, Tottori University, Yonago, Japan

J. Iga

Department of Neuropsychiatry, Ehime University Graduate School of Medicine, Ehime University, Ehime, Japan

M. Takebayashi

Department of Psychiatry and Neuroscience, Center for Metabolic Regulation of Healthy Aging, Faculty of Life Sciences, Kumamoto University, Kumamoto, Japan

measurement methods across the research sites were described previously [14]. A total of 11,955 individuals consented to participate in the study between 2016 and 2018. After excluding 55 participants who did not consent to the follow-up survey, 545 who were younger than 65 years old, 966 who had dementia at baseline, 201 without available follow-up data for incident dementia and death, and 1,854 with missing data for candidate predictors commonly available in the primary care setting, the remaining 8,334 participants (4,716 women and 3,618 men) were included in the analysis for development of Model 1, a prediction model based on predictors commonly available in the primary care setting (Supplementary Fig. 1). In addition, after excluding 21 participants with missing Geriatric Depression Scale (GDS) or Mini-Mental State Examination (MMSE) scores, the remaining 8,313 participants were included in the analysis for development of Model 2, an extended prediction model created by adding depressive symptoms and low MMSE score to Model 1. Furthermore, for development of Model 3, which extended Model 2 by adding apolipoprotein E (APOE)- ϵ 4 carrier status and brain MRI markers, 7,633 participants were included after excluding those with missing APOE- ϵ 4 genotype data ($n=130$), without available brain MRI data ($n=428$), with FreeSurfer analysis errors ($n=9$), and with outlier imaging values ($n=113$), including outliers for estimated total intracranial volume (eTIV) or more than 5 outlier regional volumes ($n=107$), and outliers for estimated hippocampal grey matter volume ($n=6$).

The present study was conducted in accordance with the Declaration of Helsinki and was approved through a centralized ethics review by the Kyushu University Institutional Board for Clinical Research (approval no. 24116). Additional site-specific approvals were obtained from each institution. We obtained written informed consent from all participants. This study followed the Transparent Reporting of a Multi-variable Prediction Model for Individual Prognosis or Diagnosis (TRIPOD) reporting guideline [15].

Follow-up survey

The detailed method of the follow-up survey is described in the Supplementary Methods. Briefly, participants were followed up prospectively from the baseline examination (2016–2018) until March 2024

using annual mail and/or telephone surveys to obtain information on vital status, cognitive impairment, and incident cardiovascular disease across all research sites. In addition, regular health examinations were conducted every 1 or 2 years at some research sites (Hirosaki City, Nakajima Town, and Hisayama Town). To accurately ascertain cognitive status, all participants who were alive at the time of the 2021–2023 survey underwent detailed face-to-face cognitive assessments. Assessments were performed using the same procedures as at baseline, either at examination sites or through visits to participants' homes or care facilities. For participants who died during follow-up, we also tried to collect death certificates and medical records as comprehensively as possible in order to obtain information on the cause of death and the presence of dementia. In the present study, the primary outcome was the composite endpoint of incident dementia or death. Among 10,933 participants included in the present analysis, outcome status for the composite endpoint during follow-up was ascertained for 10,728 participants (follow-up rate: 98.1%).

Diagnosis of dementia

The diagnostic procedure for dementia has been reported previously [14]. Dementia was diagnosed using the Diagnostic and Statistical Manual of Mental Disorders, Third Edition, Revised (DSM-III-R) [16]. A standardized two-stage procedure was used across the eight study sites [14]. In the first stage, trained physicians, public health nurses, nurses, and clinical psychologists administered the MMSE [17] as the initial screen. Individuals who met any of the following predefined criteria for possible cognitive impairment proceeded to the second stage: [i] MMSE <27 ; [ii] ≤ 4 out of 6 on the MMSE delayed-recall component [three items scored 2 points when recalled without a cue, 1 with a cue, and 0 if not recalled]; [iii] failure on the MMSE intersecting-pentagons task and/or the cube-copying test [18]; or [iv] clinical suspicion based on speech and behavior. The second-stage evaluation was conducted by expert psychiatrists or neurologists, who determined the presence of mild cognitive impairment or dementia and its subtype based on clinical examination, cognitive assessment (including the Logical Memory IIA subtest of the Wechsler Memory Scale–Revised [19] and the Pareidolia test

[20]), interviews with informants, medical records, and brain imaging. To ensure consistent diagnostic accuracy across the eight research sites, all cognitive impairment cases were adjudicated by independent expert psychiatrists or neurologists at the different recruitment sites, based on a review of the collected clinical information. Diagnoses were confirmed when the site specialists and the endpoint adjudication committee reached concordance; if they differed, the committee convened and established the final diagnosis through discussion.

Measurements of potential predictors

All information on potential predictors was obtained at the baseline survey conducted between 2016 and 2018. Participants completed a self-administered questionnaire covering education, medical history, alcohol intake, smoking status, and physical activity. Educational level was dichotomized at 9 years (≤ 9 vs. > 9 years), because 9 years is the duration of compulsory education in Japan [21]. Blood pressure was measured three times with a sphygmomanometer at intervals of ≥ 5 minutes, and the mean of the three readings was used for analysis. Hypertension was defined as blood pressure $\geq 140/90$ mmHg or current use of antihypertensive agents. Self-reported history of heart disease included prior myocardial infarction, coronary intervention, or chronic heart failure. Diabetes mellitus was defined in accordance with the 2010 American Diabetes Association criteria [22] as fasting plasma glucose ≥ 7.0 mmol/L, random plasma glucose ≥ 11.1 mmol/L, HbA1c $\geq 6.5\%$, or current use of antidiabetic medication. Serum total cholesterol was measured by an enzymatic assay. Body mass index (BMI, kg/m^2) was calculated and used to classify obesity (BMI ≥ 25.0) and leanness (BMI < 18.5). Regular exercise was defined as physical activity performed for ≥ 30 minutes at least twice weekly during the preceding year. Alcohol consumption and smoking were classified as current habitual use or not. Social interaction was assessed using the question “How often do you interact with relatives or friends who do not live with you?” Response options were “rarely,” “a few times per month,” “a few times per week,” and “almost every day.” Low social interaction was defined as a response of “rarely.” Based on a previous report indicating that sitting for 8 or more hours per day is associated with an increased risk of

mortality [23], sedentariness was defined as sitting time of ≥ 8 hours per day. APOE- $\epsilon 4$ carrier status was determined from rs429358 and rs7412 genotypes using a multiplex PCR-based targeted sequencing method [24]. Depression was diagnosed according to the Diagnostic and Statistical Manual of Mental Disorders, fourth edition [25]. Depressive symptoms were defined as a score of ≥ 6 on the GDS-short version [26]. A low MMSE score was defined as ≤ 27 points. This cutoff was chosen because the aim of the model was to predict incident dementia rather than identify prevalent cognitive impairment at baseline, and a higher threshold may better capture subtle cognitive vulnerability. A previous study also suggested that MMSE cutoffs of 27/28 may be useful for detecting mild cognitive impairment in older adults [27].

MRI analysis

Brain MRI was conducted at baseline between 2016 and 2018 using a standardized method across sites, as described previously [14, 28]. MRI equipment was configured and T1-weighted imaging (T1WI) parameters were set in accordance with the Alzheimer’s Disease Neuroimaging Initiative (ADNI) brain MRI protocols [29] at all study sites. Brain MRI was performed at each participating site using different scanners under a harmonized acquisition protocol; details of the MRI scanners and acquisition protocols at each site are provided in Supplementary Table 1. Segmentation and volumetric measurements of cortical and subcortical brain structures, including hippocampal grey matter volume and estimated total intracranial volume (eTIV), were performed automatically using FreeSurfer software (<https://surfer.nmr.mgh.harvard.edu>), version 5.3. Total brain volume was calculated from brain segment volumes excluding ventricles. Cortical segmentation was performed using the Desikan–Killiany atlas [30]. White matter lesion (WML) volumes were estimated from T1-weighted MRI using a convolutional neural network trained via cross-domain transfer learning from 2D FLAIR images, as described in Thyreau et al. [31]. We selected hippocampal grey matter volume and WML volume as brain MRI markers based on prior studies reporting their associations with dementia and mortality [32–34]. To adjust for the head size of each participant, hippocampal grey matter volume and WML volume were calculated as a percentage of eTIV.

Statistical analysis

All available data in the database were used to maximize the statistical power and generalizability of the results. We developed three prediction models with increasing levels of clinical information. Model 1 was constructed using multivariable logistic regression analysis with a backward elimination (retention criterion: $p < 0.05$ based on the Wald test), starting from a full model that included age groups (65–69, 70–74, 75–79, 80–84, and ≥ 85 years), sex (women vs. men), educational level (≤ 9 vs. > 9 years), hypertension (yes vs. no), diabetes mellitus (yes vs. no), use of lipid-lowering agents (yes vs. no), leanness (yes vs. no), obesity (yes vs. no), atrial fibrillation (yes vs. no), history of stroke (yes vs. no), history of heart disease (yes vs. no), history of respiratory disease (yes vs. no), history of liver disease (yes vs. no), history of cancer (yes vs. no), history of traumatic head injury (yes vs. no), smoking habits (current vs. past/non-smoker), drinking habits (current vs. past/non-drinker), regular exercise habit (no vs. yes), sedentariness (yes vs. no), living alone (yes vs. no), and low frequency of social interaction (yes vs. no). These potential predictors were selected based on the authors' discussion and met the following conditions: (1) they had been reported to be associated with risk of dementia or mortality, and (2) they could be easily obtained in a primary care setting. Model 2 was developed by adding depressive symptoms (yes vs. no) and low MMSE score (yes vs. no) to Model 1. Considering the increasing availability of MRI and the growing use of MRI-based health screening in Japan [35], as well as the potential value of incorporating brain imaging and genetic information into risk prediction, Model 3 further included APOE- $\epsilon 4$ carrier status (yes vs. no) and neuroimaging markers derived from brain MRI (hippocampal grey matter volume and WML volume, as continuous variables). WML volume was log-transformed to reduce skewness.

The probability of experiencing the composite endpoint (incident dementia or death) within 6.8 years was calculated as follows:

$$p = \frac{1}{1 + \exp^{-X\beta}}$$

where $X\beta$ was calculated as the intercept plus the sum of the β coefficients for each predictor multiplied

by its value (1 if present and 0 if absent for binary variables, and the observed value for continuous variables). In Models 1 and 2, which included only binary variables as predictors, a simplified risk score was also constructed by dividing the β coefficient of each predictor by the β coefficient with the smallest absolute value in each model, and rounding the resulting values to the nearest whole number [36]. The smallest absolute values of the β coefficient in Models 1 and 2 were 0.193 and 0.163, respectively. The estimated 6.8-year probability based on the simplified scores was calculated as follows:

$$p = \frac{1}{1 + \exp^{-(\text{totalscore} \times \text{smallest} \beta \text{coefficient} + \text{intercept})}}$$

The estimated 5-year probabilities based on the simplified scores were calculated using the formula $1 - (1 - P)^{5/6.8}$, where P is the 6.8-year probability.

The discrimination of the multivariable models and the simplified scores was assessed using C-statistics. For internal validation, the optimism of the developed models was estimated using the 200 bootstrap samples, and optimism-corrected C-statistics were calculated [37]. Calibration was assessed graphically by plotting the average predicted 6.8-year probabilities (based on the multivariable models and the simplified scores) against the observed 6.8-year probabilities across deciles of predicted probabilities. Calibration was also assessed quantitatively using the Hosmer–Lemeshow (HL) test, with participants grouped into deciles based on predicted probability. Decision curve analysis was performed to evaluate the potential clinical utility of the prediction models by quantifying the net benefit across a range of threshold probabilities. Net benefit was calculated by weighting true-positive and false-positive classifications according to the odds corresponding to each threshold probability [38]. The models were compared with default strategies of treating all individuals or treating none. The software package SAS version 9.4 (SAS Institute Inc., Cary, NC, USA) was used to perform all statistical analyses, and statistical significance was set at a two-tailed p value of < 0.05 in all analyses.

Results

The baseline characteristics of the participants are shown in Table 1. The median age was 72.0 years

(interquartile range, 68.0–77.0 years) and 56.6% of participants were female. Compared with participants included in the study, those excluded because of missing follow-up data or missing potential predictor data for the model in the primary care setting were older, had lower educational attainment, and were more likely to have diabetes mellitus and a history of heart disease, with standardized differences of 0.42, 0.51, 0.30, and 0.16, respectively (Supplementary Table 2).

During a median follow-up of 6.8 years, 1,151 participants (551 women and 600 men) experienced the composite endpoint of incident dementia or death. The components were 448 incident dementia cases and 801 deaths; 98 participants experienced both events during follow-up. First, we developed a multivariable prediction model for the composite endpoint of incident dementia or death using predictors commonly available in the primary care setting (Model 1). In this analysis, the predictors were selected from the full model described in the **Statistical analysis** section using multivariable logistic regression with backward elimination. Supplementary Table 3 shows the unadjusted odds ratios (ORs) and β coefficients for the associations between each potential predictor and the composite endpoint in the full model. Using backward elimination, Model 1 identified age, male sex, ≤ 9 years of formal education, diabetes mellitus, use of lipid-lowering agents, leanness, history of stroke, history of heart disease, history of respiratory disease, history of cancer, current smoking, no regular exercise habit, and low frequency of social interactions as significant predictors. The ORs and β coefficients for predictors in Model 1 are shown in Table 2. The developed prediction model was as follows: $X\beta = -3.619 + 0.776 \times (\text{age } 70\text{--}74) + 1.425 \times (\text{age } 75\text{--}79) + 2.269 \times (\text{age } 80\text{--}84) + 3.108 \times (\text{age } \geq 85) + 0.390 \times (\text{male}) + 0.278 \times (\text{education } \leq 9 \text{ years}) + 0.356 \times (\text{diabetes mellitus}) - 0.358 \times (\text{use of lipid-lowering agents}) + 0.771 \times (\text{leanness}) + 0.347 \times (\text{history of stroke}) + 0.359 \times (\text{history of heart disease}) + 0.280 \times (\text{history of respiratory disease}) + 0.232 \times (\text{history of cancer}) + 0.477 \times (\text{current smoking}) + 0.193 \times (\text{no regular exercise}) + 0.333 \times (\text{low frequency of social interactions})$. The predictive ability of Model 1, as assessed by the C-statistic, was 0.780 (95% confidence interval [CI] 0.766–0.795), and the optimism-corrected C-statistic was 0.775.

In addition, we developed a model incorporating depressive symptoms and low MMSE score to

Model 1 (Model 2; Table 3). We further extended the model by adding APOE- $\epsilon 4$ carrier status and brain MRI markers (hippocampal grey matter volume/eTIV [%] and log of [WML volume/eTIV] [%]) (Model 3; Table 4). The C-statistic and the optimism-corrected C-statistic were 0.788 (95% CI 0.773–0.802) and 0.784 for Model 2 and 0.808 (95% CI 0.794–0.823) and 0.804 for Model 3, respectively.

Calibration plots showed that the model-predicted probability and the observed probability were highly correlated (Fig. 1), and the HL tests indicated a good calibration of the developed models ($p = 0.88$, 0.56, and 0.69 for Models 1, 2, and 3, respectively). We also evaluated the predictive performance of the developed models for incident dementia and death separately. As shown in Supplementary Figures 2 and 3, the developed models showed acceptable predictive performance, with good calibration and discrimination, except for the calibration of incident dementia in Model 2 (HL test $p=0.01$).

In addition, we derived simplified risk scoring tools based on Models 1 and 2, both of which used categorical age groups and other binary predictors, to facilitate practical use. The total scores ranged from 0 to 36 and 0 to 47 in Models 1 and 2, respectively. As shown in Figure 2, the estimated 5-year probabilities for the composite endpoint increased gradually with higher total simplified risk scores in Models 1 and 2. For example, in Model 1, the estimated 5-year probabilities were 4.9%, 11.7%, and 25.1% for total scores of 5, 10, and 15, respectively; in Model 2, the corresponding probabilities were 3.5%, 7.5%, and 15.3%. The specific probabilities for each total score are also provided in Supplementary Table 4. The simplified scores based on Models 1 and 2 also showed good discrimination ability, with C-statistics of 0.779 (95% CI 0.765–0.794) and 0.786 (95% CI 0.772–0.801), respectively. The predicted probability according to the simplified scores also showed a highly linear correlation with the observed probability (Fig. 3). Supplementary Figure 4 and Supplementary Table 4 show the estimated 5-year probability of incident dementia and death based on the simplified scores derived from Models 1 and 2. Worked examples of the simplified scoring systems are presented in Supplementary Table 5. Example A represents a man with several risk factors, including shorter educational attainment, diabetes, leanness, a history of heart disease, and no exercise habit. In contrast,

Table 1 Baseline characteristics of study participants, 2016–2018

Variables	Median (IQR) or frequency
Age, years	72.0 (68.0–77.0)
65–69 years, %	37.7
70–74 years, %	27.1
75–79 years, %	19.4
80–84 years, %	10.8
≥85 years, %	5.1
Women, %	56.6
Years of education	
≤6 years, %	1.4
7–9 years, %	27.0
10–12 years, %	49.3
≥13 years, %	22.3
Systolic blood pressure, mmHg	138.7 (127.3–151.7)
Diastolic blood pressure, mmHg	78.0 (70.7–85.7)
Use of antihypertensive agents, %	50.4
Hypertension, %	72.8
Diabetes mellitus, %	16.2
Use of glucose-lowering agents, %	12.2
Serum total cholesterol, mg/dL	206.0 (183.0–230.0)
Hyperlipidemia, %	43.8
Use of lipid-lowering agents, %	29.7
Body mass index, kg/m ²	23.2 (21.1–25.3)
Leanness (body mass index <18.5 kg/m ²), %	5.5
Obesity (body mass index ≥25.0 kg/m ²), %	28.3
Atrial fibrillation, %	3.0
History of stroke, %	5.6
History of heart disease, %	5.4
History of respiratory disease, %	11.0
History of liver disease, %	5.8
History of cancer, %	13.2
History of traumatic head injury, %	4.1
Current smoking, %	8.3
Current drinking, %	43.6
Regular exercise, %	42.6
Sedentary time ≥8 hours, %	14.7
Living alone, %	14.8
Low frequency of social interactions, %	6.8
GDS score, points	2.0 (1.0–4.0)
Depressive symptoms (GDS ≥6 points), %	14.3
Depression, %	1.1
MMSE score, points	28.0 (26.0–29.0)
MMSE score ≤27 points, %	42.0
APOE-ε4 carrier, %	17.8
Hippocampal grey matter volume/eTIV, %	0.5 (0.5–0.6)
White matter lesion volume/eTIV, %	0.4 (0.2–0.9)

Abbreviation: IQR, interquartile range; GDS, Geriatric Depression Scale; MMSE, Mini-Mental State Examination; APOE, apolipoprotein E; eTIV, estimated total intracranial volume

Table 2 Model 1: Multivariable prediction model for the composite endpoint of incident dementia or death using predictors commonly available in the primary care setting

Variables ^{a)}	(Reference)	Odds ratio (95% CI)	p value	β coefficient (SE)	Score
Intercept				−3.619	
Age groups					
70–74 years	(vs. 65–69 years)	2.17 (1.74–2.71)	<0.001	0.776 (0.11)	4
75–79 years		4.16 (3.34–5.18)	<0.001	1.425 (0.11)	7
80–84 years		9.67 (7.66–12.21)	<0.001	2.269 (0.12)	12
≥85 years		22.38 (17.10–29.29)	<0.001	3.108 (0.14)	16
Men	(vs. women)	1.48 (1.28–1.71)	<0.001	0.390 (0.07)	2
Education ≤9 years	(vs. >9 years)	1.32 (1.14–1.53)	<0.001	0.278 (0.07)	1
Diabetes mellitus	(vs. no)	1.43 (1.20–1.70)	<0.001	0.356 (0.09)	2
Use of lipid-lowering agents	(vs. no)	0.70 (0.60–0.82)	<0.001	−0.358 (0.08)	−2
Leanness	(vs. no)	2.16 (1.68–2.79)	<0.001	0.771 (0.13)	4
History of stroke	(vs. no)	1.42 (1.10–1.82)	0.007	0.347 (0.13)	2
History of heart disease	(vs. no)	1.43 (1.11–1.84)	0.005	0.359 (0.13)	2
History of respiratory disease	(vs. no)	1.32 (1.08–1.62)	0.007	0.280 (0.10)	1
History of cancer	(vs. no)	1.26 (1.05–1.52)	0.01	0.232 (0.09)	1
Current smoking	(vs. no)	1.61 (1.26–2.06)	<0.001	0.477 (0.12)	2
No regular exercise	(vs. yes)	1.21 (1.05–1.40)	0.007	0.193 (0.07)	1
Low frequency of social interactions	(vs. no)	1.39 (1.10–1.78)	0.007	0.333 (0.12)	2
<i>Maximum possible total score</i>					36
<i>Index of the predictive ability for the multivariable model</i>					
C-statistic (95% CI)				0.780 (0.766–0.795)	
Optimism-corrected C-statistic				0.775	

Abbreviations: SE, standard error; CI, confidence interval

The optimism-corrected C-statistic was calculated based on 200 bootstrap samples

a) Variables were selected using a multivariable logistic regression analysis with backward elimination (retention criterion: $p < 0.05$) starting from a full model that included age group (65–69, 70–74, 75–79, 80–84, and ≥85 years), sex (women vs. men), educational level (≤9 vs. >9 years), hypertension (yes vs. no), diabetes mellitus (yes vs. no), use of lipid-lowering agents (yes vs. no), leanness (yes vs. no), obesity (yes vs. no), atrial fibrillation (yes vs. no), history of stroke (yes vs. no), history of heart disease (yes vs. no), history of respiratory disease (yes vs. no), history of liver disease (yes vs. no), history of cancer (yes vs. no), history of traumatic head injury (yes vs. no), smoking status (current vs. past/non-smoker), drinking status (current vs. past/non-drinker), regular exercise (no vs. yes), sedentary behavior (yes vs. no), living alone (yes vs. no), and low frequency of social interaction (yes vs. no).

Example B represents a man with higher-risk profile, including older age, smoking, low frequency of social interactions, depressive symptoms, and an MMSE score ≤27. Based on the simplified scoring systems, the predicted 5-year probabilities of the composite endpoint for Example A were 16.1% in Model 1 and 11.6% in Model 2, whereas those for Example B were 49.5% and 60.0%, respectively.

Finally, as shown in Supplementary Figure 5, decision curve analysis showed that Models 1, 2, and 3 provided higher net benefit than the treat-all and treat-none strategies across a range of clinically relevant threshold probabilities.

Discussion

In this study, we developed risk prediction models and simplified scoring systems for predicting the composite endpoint of incident dementia or death using longitudinal data from community-dwelling older Japanese adults. In aging societies, it has become increasingly important to shift from mere longevity to prolonging survival with preserved cognitive independence. Accordingly, we developed a set of three risk prediction models for the composite endpoint of incident dementia or death, each applicable to clinical settings with different levels of available information.

Table 3 Multivariable prediction model for the composite endpoint of incident dementia or death, extending Model 1 by adding cognitive function and depressive symptoms (Model 2)

Variables	(Reference)	Odds ratio (95% CI)	p value	β coefficient (SE)	Score
Intercept				−3.821	
Age groups					
70–74 years	(vs. 65–69 years)	2.13 (1.71–2.67)	<0.001	0.757 (0.11)	5
75–79 years		4.00 (3.20–4.99)	<0.001	1.385 (0.11)	8
80–84 years		8.89 (7.03–11.26)	<0.001	2.185 (0.12)	13
≥85 years		20.70 (15.76–27.18)	<0.001	3.030 (0.14)	19
Men	(vs. women)	1.44 (1.25–1.67)	<0.001	0.368 (0.08)	2
Education ≤9 years	(vs. >9 years)	1.18 (1.01–1.37)	0.03	0.163 (0.08)	1
Diabetes mellitus	(vs. no)	1.40 (1.17–1.66)	<0.001	0.333 (0.09)	2
Use of lipid-lowering agents	(vs. no)	0.71 (0.61–0.84)	<0.001	−0.339 (0.08)	−2
Leanness	(vs. no)	2.13 (1.64–2.75)	<0.001	0.754 (0.13)	5
History of stroke	(vs. no)	1.39 (1.08–1.79)	0.01	0.331 (0.13)	2
History of heart disease	(vs. no)	1.37 (1.06–1.77)	0.02	0.315 (0.13)	2
History of respiratory disease	(vs. no)	1.34 (1.10–1.65)	0.005	0.294 (0.10)	2
History of cancer	(vs. no)	1.25 (1.04–1.51)	0.02	0.225 (0.09)	1
Current smoking	(vs. no)	1.53 (1.20–1.97)	<0.001	0.428 (0.13)	3
No regular exercise	(vs. yes)	1.19 (1.03–1.37)	0.02	0.174 (0.07)	1
Low frequency of social interactions	(vs. no)	1.29 (1.01–1.65)	0.04	0.255 (0.12)	2
Depressive symptoms (GDS ≥6 points)	(vs. no)	1.34 (1.12–1.60)	0.002	0.290 (0.09)	2
MMSE score ≤27 points	(vs. ≥28 points)	1.69 (1.46–1.94)	<0.001	0.522 (0.07)	3
<i>Maximum possible total score</i>					47
<i>Index of the predictive ability for the multivariable model</i>					
C-statistic (95% CI)				0.788 (0.773–0.802)	
Optimism-corrected C-statistic				0.784	

Abbreviations: SE, standard error; CI, confidence interval; GDS, Geriatric Depression Scale; MMSE, Mini-Mental State Examination

The optimism-corrected C-statistic was calculated based on 200 bootstrap samples

These risk prediction tools showed good discrimination and calibration for the composite endpoint, as well as for incident dementia and death. The decision curve analysis suggested that the models may provide clinical utility across a range of threshold probabilities and may therefore support risk-informed decision-making in practice. Evidence from a community-based prospective cohort study of 2,449 participants indicated that adherence to multiple healthy lifestyle factors was associated not only with longer overall life expectancy but also with increased life expectancy without Alzheimer's dementia [39]. This finding supports the rationale for the models developed in the present study. These models may aid risk stratification and serve as a guide for interventions aimed at prolonging dementia-free survival at the individual level. For example, individuals at higher

predicted risk may warrant closer clinical monitoring and more intensive management of modifiable risk factors and lifestyle-related conditions.

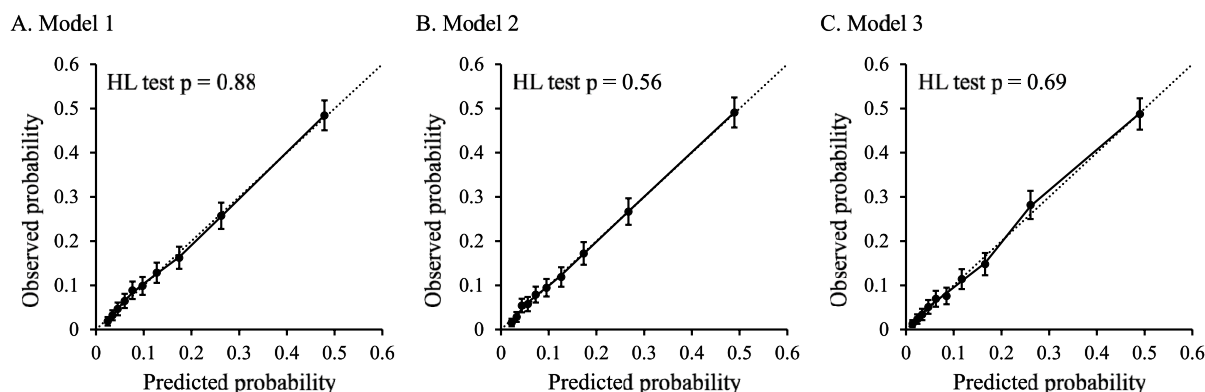
Two previous studies have reported prediction models for disability-free survival, an outcome similar to the present study [40, 41]. The model developed from community-dwelling participants in the Aspirin in Reducing Events in the Elderly trial included cognitive function and gait speed as key predictors [40]. Similarly, the model derived from the community-based Swedish National Study on Aging and Care in Kungsholmen cohort incorporated high cognitive performance, high life satisfaction, non-smoking, engagement in physical leisure activity, and absence of chronic diseases [41]. Consistent with these studies, our models also retained lifestyle factors, prevalent chronic conditions, and neurocognitive measures

Table 4 Multivariable prediction model for the composite endpoint of incident dementia or death, extending Model 2 by adding APOE-ε4 carrier status and brain MRI markers (Model 3)

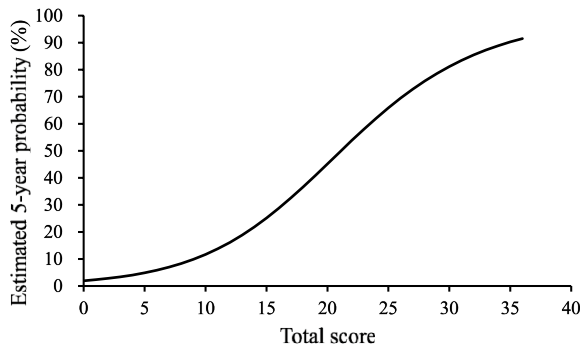
Variables	(Reference)	Odds ratio (95% CI)	p value	β coefficient (SE)
Intercept				0.104
Age groups				
70–74 years	(vs. 65–69 years)	1.64 (1.29–2.09)	<0.001	0.495 (0.12)
75–79 years		2.36 (1.84–3.03)	<0.001	0.859 (0.13)
80–84 years		4.05 (3.07–5.35)	<0.001	1.399 (0.14)
≥85 years		7.92 (5.67–11.06)	<0.001	2.069 (0.17)
Men	(vs. women)	1.11 (0.94–1.31)	0.23	0.103 (0.09)
Education ≤9 years	(vs. >9 years)	1.15 (0.98–1.35)	0.09	0.140 (0.08)
Diabetes mellitus	(vs. no)	1.41 (1.17–1.70)	<0.001	0.341 (0.10)
Use of lipid-lowering agents	(vs. no)	0.72 (0.61–0.85)	<0.001	−0.330 (0.09)
Leanness	(vs. no)	2.08 (1.57–2.77)	<0.001	0.734 (0.15)
History of stroke	(vs. no)	1.10 (0.83–1.47)	0.51	0.098 (0.15)
History of heart disease	(vs. no)	1.26 (0.94–1.67)	0.12	0.227 (0.15)
History of respiratory disease	(vs. no)	1.39 (1.12–1.74)	0.003	0.330 (0.11)
History of cancer	(vs. no)	1.23 (1.01–1.50)	0.04	0.204 (0.10)
Current smoking	(vs. no)	1.47 (1.13–1.91)	0.005	0.383 (0.14)
No regular exercise	(vs. yes)	1.14 (0.98–1.33)	0.10	0.128 (0.08)
Low frequency of social interactions	(vs. no)	1.22 (0.94–1.60)	0.13	0.202 (0.14)
Depressive symptoms (GDS ≥6 points)	(vs. no)	1.29 (1.06–1.57)	0.01	0.257 (0.10)
MMSE score ≤27 points	(vs. ≥28 points)	1.54 (1.32–1.80)	<0.001	0.432 (0.08)
APOE-ε4 carrier	(vs. no)	1.23 (1.01–1.49)	0.04	0.203 (0.10)
Hippocampal grey matter volume/eTIV (%)	(per 0.1 % decrease)	1.93 (1.70–2.19)	<0.001	0.659 (0.06)
White matter lesion volume/eTIV (%), log-transformed	(per 1 log [%] unit increase)	1.27 (1.17–1.38)	<0.001	0.241 (0.04)
<i>Index of the predictive ability for the multivariable model</i>				
C-statistic (95% CI)				0.808 (0.794–0.823)
Optimism-corrected C-statistic				0.804

Abbreviations: SE, standard error; CI, confidence interval; GDS, Geriatric Depression Scale; MMSE, Mini-Mental State Examination; APOE, apolipoprotein E; eTIV, estimated total intracranial volume.

The optimism-corrected C-statistic was calculated based on 200 bootstrap samples.

**Fig. 1** Calibration plots for predicting the composite endpoint of incident dementia or death. The predicted probability was estimated from the multivariable models. The solid curve is the Loess-estimated calibration curve. HL test, Hosmer–Lemeshow test

A. Model 1



B. Model 2

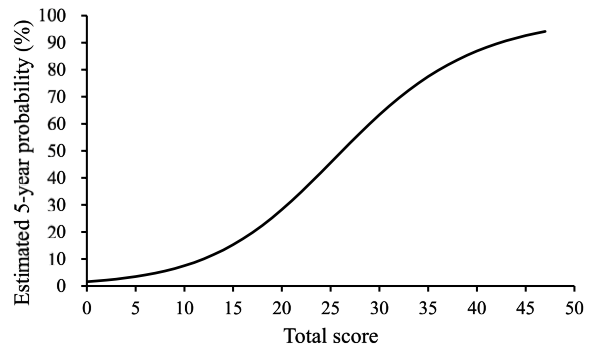
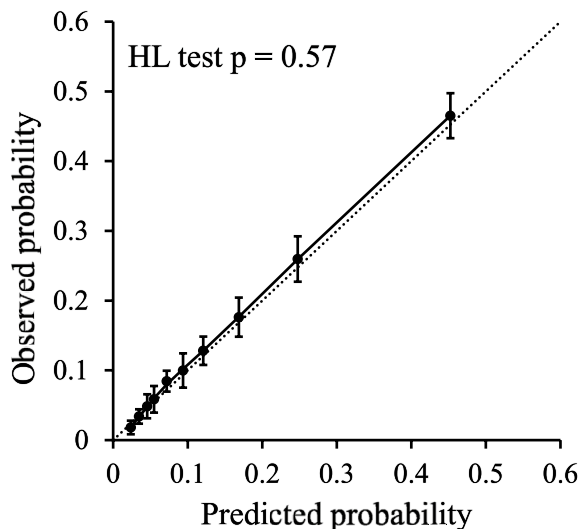


Fig. 2 Estimated 5-year probability of the composite endpoint of incident dementia or death based on simplified scores

A. Model 1, simplified score



B. Model 2, simplified score

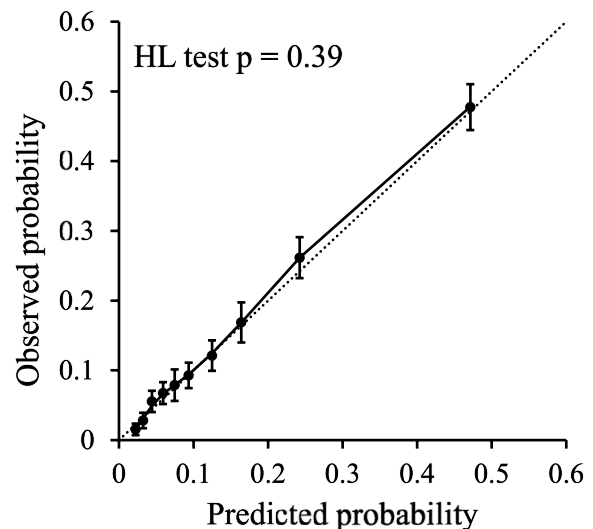


Fig. 3 Calibration plots for predicting the composite endpoint of incident dementia or death. The predicted probability was estimated from the simplified score. The solid curve is the Loess-estimated calibration curve. HL test, Hosmer–Lemeshow test

as predictors. In contrast to previous studies, we developed a set of prediction models based on the information available in different clinical settings, which may improve practical applicability in real-world practice.

The prediction models we developed incorporated well-established risk factors for incident dementia and death. Older age, diabetes mellitus, and physical inactivity have consistently been reported to be associated with higher risks of both incident dementia and mortality [42]. With regard to sex, a recent meta-analysis has reported that incidence of dementia is similar

in women and men [43], but men consistently exhibit higher mortality [1]. Therefore, the finding that men had a higher risk of the composite endpoint in our model is plausible. In addition, our models included use of lipid-lowering medication. Meta-analyses of observational studies reported that statins—the most commonly prescribed lipid-lowering agents [44]—have been associated with lower risks of both incident dementia and mortality [45, 46], which supports our results. Low frequency of social interaction was also included among the model's predictors. Previous meta-analyses have shown that social isolation is

associated with higher mortality [47], whereas good social engagement is associated with a lower risk of incident dementia [48].

In the present study, the risk prediction model that incorporated depressive symptoms, MMSE score, APOE- ϵ 4 carrier status, and brain MRI data (Models 2 and 3) further improved the predictive performance. Given that the MMSE is a well-established measure of cognitive function and late-life depression may be prodromal to neurodegenerative disease [49], it is reasonable that incorporating MMSE and depressive symptoms improved the predictive performance. With regard to brain MRI markers, community-based prospective cohorts reported that smaller hippocampal volumes were associated with increased risk of incident dementia [32]. Furthermore, meta-analyses indicate that greater WML volume is associated with higher risks of both dementia and mortality [34]. The APOE- ϵ 4 allele is well known as a risk factor for both dementia and mortality [50, 51]. Taken together, the predictors retained in our models are concordant with prior evidence and are considered reasonable. In addition, even when predicting each endpoint separately, these models showed generally good performance for incident dementia and mortality. As discussed above, incorporating more detailed evaluations improved the model's predictive performance. In real-world clinical settings, it is important to select an appropriate model according to the available resources.

The strengths of this study include its large sample size and the development of prediction models tailored to the different levels of information available across diverse clinical settings. However, several limitations should be noted. First, we did not externally validate the developed models; therefore, their generalizability beyond community-dwelling older adults in Japan remains uncertain. When applied to populations with different ethnic, demographic, cultural, or clinical characteristics, the models may overestimate or underestimate risk, with potential deterioration in both calibration and discrimination. However, the direction and magnitude of such bias could not be determined in the present study. Second, the use of a composite endpoint of incident dementia or death represents a methodological limitation. In this study, dementia status was assessed at two time points for most participants: the baseline examination (2016–2018) and the follow-up examination (2021–2023). Therefore, participants who

died before the follow-up examination could not always be adequately evaluated for incident dementia. We therefore adopted a composite endpoint as a pragmatic approach to account for incomplete ascertainment of dementia due to death during follow-up. Third, although we used backward elimination for variable selection, this approach may be associated with model instability and overfitting. However, in the present study, 1,151 events were observed for 24 candidate predictors, corresponding to approximately 48 events per candidate predictor, which is generally considered a setting at lower risk of serious overfitting [52]. In addition, a previous study suggested that, in large samples with sufficient information, conventional selection methods may achieve predictive performance broadly comparable to that of penalized approaches [53]. Therefore, we considered backward elimination to be an acceptable approach in the present study, although penalized or shrinkage-based approaches, such as lasso regression, may provide a more stable alternative for variable selection. Fourth, during the follow-up period, some participants may have received interventions targeting risk factors, such as management of vascular risk factors (e.g., hypertension and diabetes mellitus). These changes in treatment may have altered individual risk trajectories, attenuated the observed associations between predictors and outcomes, and slightly decreased the model's predictive performance. Fifth, individuals excluded because of unavailable follow-up data or missing predictor data for the primary care model were older, had lower educational attainment, and were more likely to have diabetes mellitus and a history of heart disease than those included; consequently, our study may have underestimated the risks of incident dementia and mortality. Sixth, because this was an observational study, the associations identified in the models should not be interpreted as causal effects, and the identified predictors should not be directly considered intervention targets. Further interventional studies are needed to determine the extent to which risk can be reduced by modifying these factors. Seventh, detailed information on the dose or intensity of alcohol consumption and cigarette smoking was not available; therefore, potential dose–response relationships could not be evaluated.

In conclusion, we developed risk prediction models and simplified scores for the composite endpoint of incident dementia or mortality among

community-dwelling older Japanese adults. Risk stratification using our models could provide useful information for prolonging cognitively healthy survival.

Acknowledgements The authors would like to acknowledge the diligent work and contributions of all researchers and investigators in the JPSC-AD Study Group. They would also like to thank the participants for their time in taking part in the JPSC-AD study. Finally, we are grateful for the use of computer resources offered under the category of General Projects by the Research Institute for Information Technology, Kyushu University, which we used to conduct the statistical analyses.

Author contributions XM, TH, and TN contributed to the study conception and study design; XM, TH, and TN contributed to the data analysis; XM, TH, TO, YF, MS, JH, YT, T. Mikami, T. Maeda, KO, MM, RH, MT, and TN contributed to the data collection; XM, TH, TO, YF, MS, JH, YT, T. Mikami, T. Maeda, KO, MM, RH, JI, MT, and TN contributed to the data interpretation. XM wrote the first draft of the manuscript and all authors contributed to critical revision of the manuscript.

Funding This study was supported by the Japan Agency for Medical Research and Development (grant nos. dk0207025 and dk0207053), Suntory Holdings, Ltd. (Osaka, Japan), and the Health and Labour Sciences Research Grants of the Ministry of Health, Labour and Welfare of Japan (JPMH24GB1002). The funders had no role in the design of the study, the collection, analysis, and interpretation of data, or the writing of the manuscript.

Data availability The datasets generated and analyzed during this study are not publicly available because of restrictions in the informed consent of the study participants. However, data may be made available on reasonable request to the principal investigator, Toshiharu Ninomiya, and with permission from the JPSC-AD steering committee and the Japan Agency for Medical Research and Development.

Declarations

Declaration of generative artificial intelligence (AI) in writing The authors only used generative AI and AI-assisted technologies to improve language usage and readability.

Conflict of interest Toshiharu Ninomiya received research grants from Suntory Holdings, Ltd., Japan.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated

otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

1. GBD 2021 Demographics Collaborators. Global age-sex-specific mortality, life expectancy, and population estimates in 204 countries and territories and 811 subnational locations, 1950–2021, and the impact of the COVID-19 pandemic: a comprehensive demographic analysis for the Global Burden of Disease Study 2021. *Lancet*. 2024;403:1989–2056. [https://doi.org/10.1016/S0140-6736\(24\)00476-8](https://doi.org/10.1016/S0140-6736(24)00476-8).
2. Xiaopeng Z, Jing Y, Xia L, et al. Global burden of Alzheimer's disease and other dementias in adults aged 65 years and older, 1991–2021: population-based study. *Front Public Health*. 2025;13:1585711. <https://doi.org/10.3389/fpubh.2025.1585711>.
3. Liang CS, Li DJ, Yang FC, et al. Mortality rates in Alzheimer's disease and non-Alzheimer's dementias: a systematic review and meta-analysis. *Lancet Healthy Longev*. 2021;2:e479–88. [https://doi.org/10.1016/S2666-7568\(21\)00140-9](https://doi.org/10.1016/S2666-7568(21)00140-9).
4. Agüero-Torres H, Fratiglioni L, Guo Z, Viitanen M, von Strauss E, Winblad B. Dementia is the major cause of functional dependence in the elderly: 3-year follow-up data from a population-based study. *Am J Public Health*. 1998;88:1452–6. <https://doi.org/10.2105/ajph.88.10.1452>.
5. Zhong S, Xiao C, Li R, Lan Y, et al. The global, regional, and national burdens of dementia in 204 countries and territories from 1990 to 2021: a trend analysis based on the Global Burden of Disease Study 2021. *Medicine (Baltimore)*. 2025;104:e41836. <https://doi.org/10.1097/MD.00000000000041836>.
6. Fried LP, Rowe JW. Health in aging - past, present, and future. *N Engl J Med*. 2020;383(14):1293–6. <https://doi.org/10.1056/NEJMp2016814>.
7. Jacob ME, Yee LM, Diehr PH, et al. Can a healthy lifestyle compress the disabled period in older adults? *J Am Geriatr Soc*. 2016;64:1952–61. <https://doi.org/10.1111/jgs.14314>.
8. Bress AP, Bellows BK, King JB, et al. Cost-effectiveness of intensive versus standard blood-pressure control. *N Engl J Med*. 2017;377:745–55. <https://doi.org/10.1056/NEJMs1616035>.
9. McRae I, Zheng L, Bourke S, Cherbuin N, Anstey KJ. Cost-effectiveness of dementia prevention interventions. *J Prev Alzheimers Dis*. 2021;8:210–7. <https://doi.org/10.14283/jpad.2020.71>.
10. Brain J, Kafadar AH, Errington L, et al. What's new in dementia risk prediction modelling? An updated systematic review. *Dement Geriatr Cogn Dis Extra*. 2024;14:49–74. <https://doi.org/10.1159/000539744>.

11. Exmann CJC, Kooijmans ECM, Joling KJ, Burchell GL, Hoogendijk EO, van Hout HPJ. Mortality prediction models for community-dwelling older adults: a systematic review. *Ageing Res Rev.* 2024;101:102525. <https://doi.org/10.1016/j.arr.2024.102525>.
12. Hou XH, Feng L, Zhang C, Cao XP, Tan L, Yu JT. Models for predicting risk of dementia: a systematic review. *J Neurol Neurosurg Psychiatry.* 2019;90:373–9. <https://doi.org/10.1136/jnnp-2018-318212>.
13. Payton NM, Kalpouzos G, Rizzuto D, et al. Combining cognitive, genetic, and structural neuroimaging markers to identify individuals with increased dementia risk. *J Alzheimers Dis.* 2018;64:533–42. <https://doi.org/10.3233/JAD-180199>.
14. Ninomiya T, Nakaji S, Maeda T, et al. Study design and baseline characteristics of a population-based prospective cohort study of dementia in Japan: the Japan Prospective Studies Collaboration for Aging and Dementia (JPSC-AD). *Environ Health Prev Med.* 2020;25:64. <https://doi.org/10.1186/s12199-020-00903-3>.
15. Collins GS, Reitsma JB, Altman DG, Moons KG. Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD): the TRIPOD statement. *BMJ.* 2015;350:g7594. <https://doi.org/10.1136/bmj.g7594>.
16. American Psychiatric Association. Diagnostic and Statistical Manual of Mental Disorders, 3rd Revised Edition (1987)
17. Folstein MF, Folstein SE, McHugh PR. "Mini-mental state". A practical method for grading the cognitive state of patients for the clinician. *J Psychiatr Res.* 1975;12:189–98. [https://doi.org/10.1016/0022-3956\(75\)90026-6](https://doi.org/10.1016/0022-3956(75)90026-6).
18. Maeshima S, Osawa A, Maeshima E, et al. Usefulness of a cube-copying test in outpatients with dementia. *Brain Inj.* 2004;18:889–98. <https://doi.org/10.1080/02699050410001671847>.
19. Procedures manual – ADNI: <https://adni.loni.usc.edu/wp-content/uploads/2008/07/adni2-procedures-manual.pdf>. Accessed 16 Oct 2025
20. Mamiya Y, Nishio Y, Watanabe H, et al. The pareidolia test: a simple neuropsychological test measuring visual hallucination-like illusions. *PLoS One.* 2016;11:e0154713. <https://doi.org/10.1371/journal.pone.0154713>.
21. Ministry of Education, Culture, Sports, Science and Technology (MEXT). Overview of the education system in Japan. MEXT. <https://www.mext.go.jp/en/policy/education/overview/index.htm>. Accessed 19 Mar 2026
22. American Diabetes Association. Diagnosis and classification of diabetes mellitus. *Diabetes Care.* 2010;33(Suppl 1):S62–9. <https://doi.org/10.2337/dc10-S062>.
23. Patterson R, McNamara E, Tainio M, et al. Sedentary behaviour and risk of all-cause, cardiovascular and cancer mortality, and incident type 2 diabetes: a systematic review and dose response meta-analysis. *Eur J Epidemiol.* 2018;33:811–29. <https://doi.org/10.1007/s10654-018-0380-1>.
24. Momozawa Y, Akiyama M, Kamatani Y, et al. Low-frequency coding variants in CETP and CFB are associated with susceptibility of exudative age-related macular degeneration in the Japanese population. *Hum Mol Genet.* 2016;25:5027–34. <https://doi.org/10.1093/hmg/ddw335>.
25. American Psychiatric Association. Diagnostic and Statistical Manual of Mental Disorders. 4th ed. Washington, DC: American Psychiatric Association; 1994.
26. Sheikh JI, Yesavage JA. Geriatric depression scale (GDS): recent evidence and development of a shorter version. *Clin Gerontol.* 1986;5:165–73. https://doi.org/10.1300/J018v05n01_09.
27. Ciesielska N, Sokołowski R, Mazur E, Podhorecka M, Polak-Szabela A, Kędziora-Kornatowska K. Is the Montreal cognitive assessment (MoCA) test better suited than the mini-mental state examination (MMSE) in mild cognitive impairment (MCI) detection among people aged over 60? Meta-analysis. *Psychiatr Pol.* 2016;50:1039–52. <https://doi.org/10.12740/PP/45368>.
28. Hirabayashi N, Honda T, Hata J, et al. Association between frequency of social contact and brain atrophy in community-dwelling older people without dementia: The JPSC-AD study. *Neurology.* 2023;101:e1108–17. <https://doi.org/10.1212/WNL.0000000000207602>.
29. Jack CR Jr, Bernstein MA, Fox NC, et al. The Alzheimer's Disease Neuroimaging Initiative (ADNI): MRI methods. *J Magn Reson Imaging.* 2008;27:685–91. <https://doi.org/10.1002/jmri.21049>.
30. Desikan RS, Ségonne F, Fischl B, et al. An automated labeling system for subdividing the human cerebral cortex on MRI scans into gyral based regions of interest. *Neuroimage.* 2006;31:968–80. <https://doi.org/10.1016/j.neuroimage.2006.01.021>.
31. Thyreau B, Tatewaki Y, Chen L, et al. Higher-resolution quantification of white matter hypointensities by large-scale transfer learning from 2D images on the JPSC-AD cohort. *Hum Brain Mapp.* 2022;43:3998–4012. <https://doi.org/10.1002/hbm.25899>.
32. Nakazawa T, Ohara T, Hirabayashi N, et al. Multiple-region grey matter atrophy as a predictor for the development of dementia in a community: The Hisayama study. *J Neurol Neurosurg Psychiatry.* 2022;93:263–71. <https://doi.org/10.1136/jnnp-2021-326611>.
33. Ikram MA, Vernooij MW, Vrooman HA, Hofman A, Breteler MM. Brain tissue volumes and small vessel disease in relation to the risk of mortality. *Neurobiol Aging.* 2009;30:450–6. <https://doi.org/10.1016/j.neurobiolaging.2007.07.009>.
34. DeBette S, Markus HS. The clinical importance of white matter hyperintensities on brain magnetic resonance imaging: systematic review and meta-analysis. *BMJ.* 2010;341:c3666. <https://doi.org/10.1136/bmj.c3666>.
35. OECD. Health at a Glance 2025: OECD Indicators. Paris: OECD Publishing; 2025. <https://doi.org/10.1787/8f9e3f98-en>.
36. Sullivan LM, Massaro JM, D'Agostino RB. Presentation of multivariate data for clinical use: the Framingham Study risk score functions. *Stat Med.* 2004;23:1631–60. <https://doi.org/10.1002/sim.1742>.
37. Harrell FE. Regression modeling strategies: with applications to linear models, logistic and ordinal regression, and survival analysis. New York: Springer; 2015.
38. Vickers AJ, Elkin EB. Decision curve analysis: a novel method for evaluating prediction models. *Med Decis Making.* 2006;26:565–74. <https://doi.org/10.1177/0272989X06295361>.

39. Dhana K, Franco OH, Ritz EM, et al. Healthy lifestyle and life expectancy with and without Alzheimer's dementia: population based cohort study. *BMJ*. 2022;377:e068390. <https://doi.org/10.1136/bmj-2021-068390>.
40. Neumann JT, Thao LTP, Murray AM, et al. Prediction of disability-free survival in healthy older people. *Geroscience*. 2022;44:1641–55. <https://doi.org/10.1007/s11357-022-00547-x>.
41. Wu W, Guo J, Dove A, Dunk MM, Xu W. A nomogram for predicting disability-free survival in older adults over 15 years: a population-based cohort study. *Arch Gerontol Geriatr*. 2024;126:105527. <https://doi.org/10.1016/j.archger.2024.105527>.
42. Livingston G, Huntley J, Liu KY, et al. Dementia prevention, intervention, and care: 2024 report of the Lancet standing Commission. *Lancet*. 2024;404:572–628. [https://doi.org/10.1016/S0140-6736\(24\)01296-0](https://doi.org/10.1016/S0140-6736(24)01296-0).
43. Huque H, Eramudugolla R, Chidiac B, et al. Could country-level factors explain sex differences in dementia incidence and prevalence? A systematic review and meta-analysis. *J Alzheimers Dis*. 2023;91:1231–41. <https://doi.org/10.3233/JAD-220724>.
44. Blais JE, Wei Y, Yap KKW, et al. Trends in lipid-modifying agent use in 83 countries. *Atherosclerosis*. 2021;328:44–51. <https://doi.org/10.1016/j.atherosclerosis.2021.05.016>.
45. Westphal Filho FL, Moss Lopes PR, Menegaz de Almeida A, et al. Statin use and dementia risk: a systematic review and updated meta-analysis. *Alzheimers Dement N Y*. 2025;11:e70039. <https://doi.org/10.1002/trc2.70039>.
46. Awad K, Mohammed M, Zaki MM, et al. Association of statin use in older people primary prevention group with risk of cardiovascular events and mortality: a systematic review and meta-analysis of observational studies. *BMC Med*. 2021;19:139. <https://doi.org/10.1186/s12916-021-02009-1>.
47. Wang F, Gao Y, Han Z, et al. A systematic review and meta-analysis of 90 cohort studies of social isolation, loneliness and mortality. *Nat Hum Behav*. 2023;7:1307–19. <https://doi.org/10.1038/s41562-023-01617-6>.
48. Wang S, Molassiotis A, Guo C, Leung ISH, Leung AYM. Association between social integration and risk of dementia: a systematic review and meta-analysis of longitudinal studies. *J Am Geriatr Soc*. 2023;71:632–45. <https://doi.org/10.1111/jgs.18094>.
49. Brain J, Alshahrani M, Kafadar AH, et al. Temporal dynamics in the association between depression and dementia: an umbrella review and meta-analysis. *EclinicalMedicine*. 2025;84:103266. <https://doi.org/10.1016/j.eclinm.2025.103266>.
50. Farrer LA, Cupples LA, Haines JL, et al. Effects of age, sex, and ethnicity on the association between apolipoprotein E genotype and Alzheimer disease. A meta-analysis. APOE and Alzheimer Disease Meta Analysis Consortium. *JAMA*. 1997;278:1349–56. <https://doi.org/10.1001/jama.1997.03550160069041>.
51. Wolters FJ, Yang Q, Biggs ML, et al. The impact of APOE genotype on survival: results of 38,537 participants from six population-based cohorts (E2-CHARGE). *PLoS One*. 2019;14:e0219668. <https://doi.org/10.1371/journal.pone.0219668>.
52. van der Ploeg T, Austin PC, Steyerberg EW. Modern modelling techniques are data hungry: a simulation study for predicting dichotomous endpoints. *BMC Med Res Methodol*. 2014;14:137. <https://doi.org/10.1186/1471-2288-14-137>.
53. Kipruto E, Sauerbrei W. Evaluating prediction performance: a simulation study comparing penalized and classical variable selection methods in low-dimensional data. *Appl Sci*. 2025;15:7443. <https://doi.org/10.3390/app15137443>.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.